

(varied process parameters) and skewed circuit boards (varied trace impedance)

Post-debug challenges

1. Basic observability is package pins
 - Signal observability (higher integration SoC)
 - Probing scope
 - Probing signal integrity
2. Trend is towards lower observability. Integration increasing towards SoC
3. Functional and circuit issues require different solutions

Observability/control/survivability architecture

Fictional example code:

```
// Read a counter in a certain design block
// on a certain boundary scan chain
// First select a desired design block
irscan(MY_CHAIN, SELECT_BLOCK_OPCODE)
// scan in block number stored in 4 bits.
drscan(MY_CHAIN, 4, BLOCK_NUMBER)
// now select the desired counter out of
// 100 counters
// 7 bits used to select counter
// This operation also resets counter to 0
irscan(MY_CHAIN, COUNTER_SELECT_OPCODE)
drscan(MY_CHAIN, 7, COUNTER_ID)
// Wait until something triggers...
// Read Counter Value
irscan(MY_CHAIN, READ_COUNTER_OPCODE)
drscan(MY_CHAIN, 32, 0x0, Counter_Value)
```

Example of a boundary scan

Depending on the deployment target, use-cases and necessary power/performance trade-offs, any design functionality in an IP may be moved to hardware or software (firmware) implementation. Systems are vertically integrated with system use-cases only realised by hardware, software, applications and peripheral communication.

Validation and post-silicon validation is a complex co-validation problem across hardware, software and peripheral functionality, with no clear decomposition into individual components. With integration of significant design functionalities into one system, it is getting more and more complex to control and observe any individual design component as necessary for validation. And with

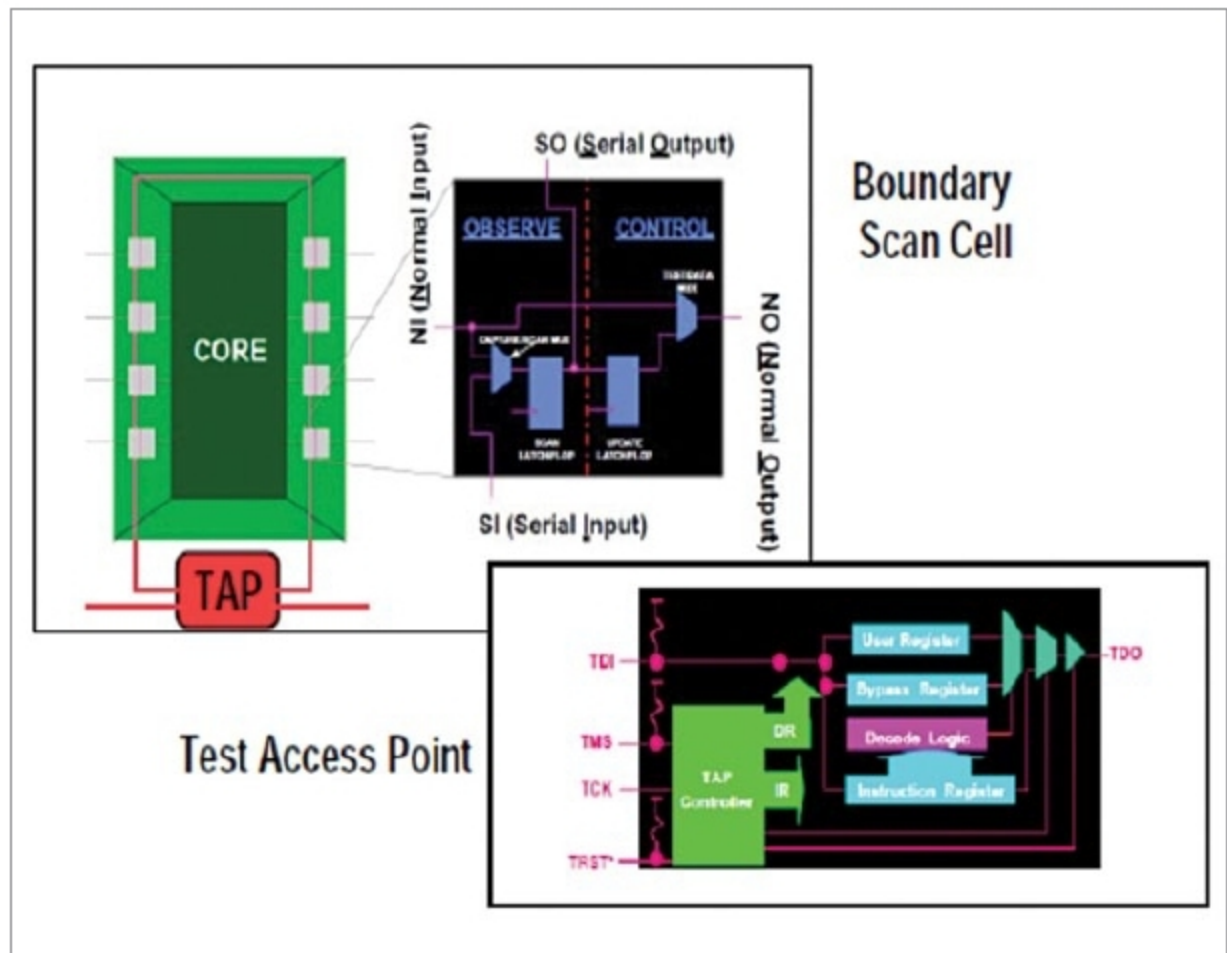


Fig. 14: Boundary scan

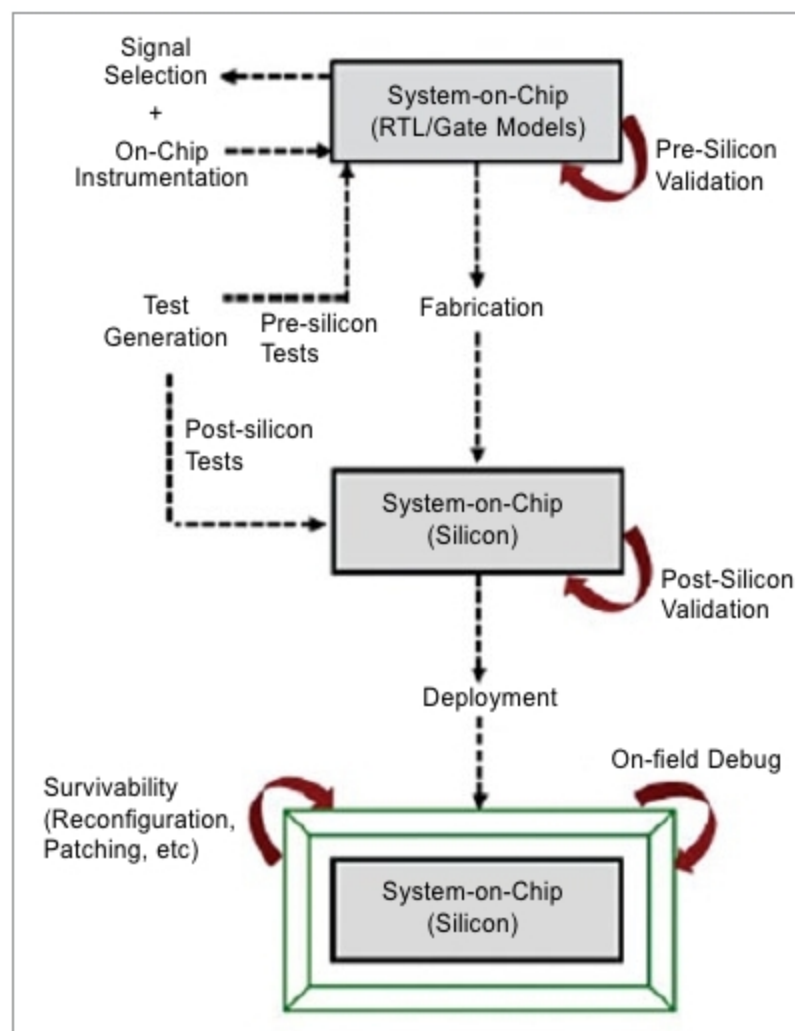


Fig. 15: Stages of SoC validation: pre-silicon validation, post-silicon validation and on-field survivability

reduced time-to-market, the number of silicon spins available for validation has decreased dramatically.

When an error is found in silicon, other errors are detected in the same silicon spin. In fact, tools, flows and

design instrumentations have been incrementally accumulated over time in response to specific challenges or requirements. Over 20 per cent of the design real estate and a significant component of the CAD flow effort are devoted towards silicon validation today.

As we move from pre-silicon to post-silicon and finally on-field execution, more and more complex usage scenarios are exercised, potentially stimulating errors that could not be seen in previous validation phases. At the same time, observability and controllability of the design during these executions get progres-

sively more complex, making it harder to root cause failure. At the same time, cost of a bug has increased and time available for debug decreased as we go further in the system lifecycle.

To be continued